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Concrete Experience

United States Army
Cyber Center of Excellence

Fundamentals of Electromagnetic (EM) Reconnaissance

112-17ABOLC

OVERALL CLASSIFICATION:

UNCLASSIFIED

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Learning Step Activities

- LSA 1 - Define the Fundamentals of Reconnaissance
- LSA 2 - Identify Key Information Reporting for Electromagnetic Spectrum (EMS) Reconnaissance
- LSA 3 - Discuss how to Gain and Maintain Enemy Contact in the EMS

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Administrative Information

Safety Requirements	NONE
Risk Assessment Level	LOW
Environmental Considerations	None
Evaluation	None

References

- FM 3-12. *Cyberspace and Electromagnetic Warfare Operations*. 17 Sep 2025.
- FM 3-98. *Reconnaissance And Security Operations*. 10 Jan 2023.
- ATP 3-12.3, C1. *Electromagnetic Warfare Techniques*. 18 Sep 2024.
- ATP 3-12.4. *Electromagnetic Warfare Platoon*. 09 Jan 2023.

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Define the Fundamentals of Reconnaissance

Reconnaissance

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- A mission undertaken to obtain information about the activities and resources of an enemy or adversary, or to secure data concerning the meteorological, hydrographic, geographic, or other characteristics of a particular area, by visual observation or other detection methods.
- Reconnaissance operations allow commanders to understand the situation, visualize the battle, and make decisions.

Reconnaissance

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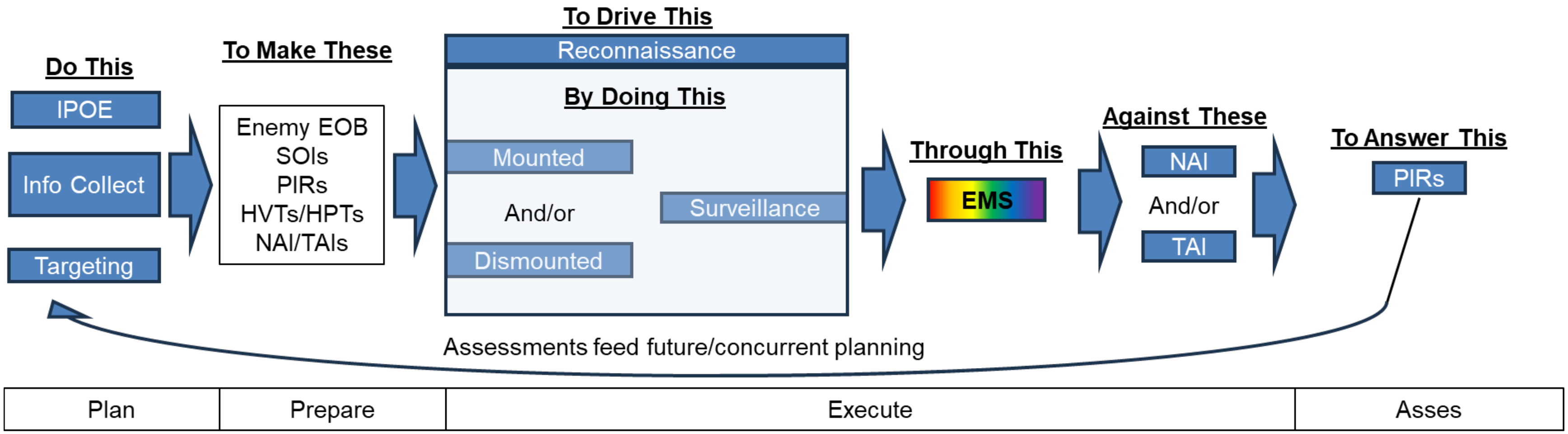
- The integrating processes for reconnaissance and security operations:
  - IPOE
  - Targeting
  - Information collection



Electromagnetic Reconnaissance Overview

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- Electromagnetic reconnaissance is the detection, location, identification, and evaluation of foreign electromagnetic radiations.
  - Electromagnetic reconnaissance is one of the most common mission sets of the EW PLT.
  - Electromagnetic reconnaissance allows EW PLTs to directly impact the BDE's operations and is a power multiplier to other concurrent ISR activities.
  - Electromagnetic reconnaissance is delicate blend of tactical and technical activities.
  - Information from electromagnetic reconnaissance can lead to the implementation or modification of EP measures or dictate EA actions.

Reconnaissance Techniques – Reconnaissance-push

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- Reconnaissance that refines the common operational picture, enabling the commander to finalize the plan and support shaping and decisive operations.
- Commanders use reconnaissance-push when they have relatively thorough understanding of the operational environment.
  - In these cases, commanders “push” reconnaissance assets into specific portions of their areas of operations to confirm, deny, and validate planning assumptions affecting operations.
- Reconnaissance-push emphasizes detailed, well-rehearsed planning.

Reconnaissance Techniques – Reconnaissance-pull

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- Reconnaissance that determines which routes are suitable for maneuver, where the enemy is strong and weak, and where gaps exist, thus pulling the main body toward and along the path of least resistance.
- Commanders use reconnaissance-pull when they are uncertain of the compositions and dispositions of enemy forces in their areas of operations, information concerning terrain is vague, and time is limited.
  - In these cases, reconnaissance assets initially work over broad areas to develop the enemy situations.
- As they gain an understanding of enemy weaknesses, they then “pull” the main body to positions of tactical advantage.

Reconnaissance Types

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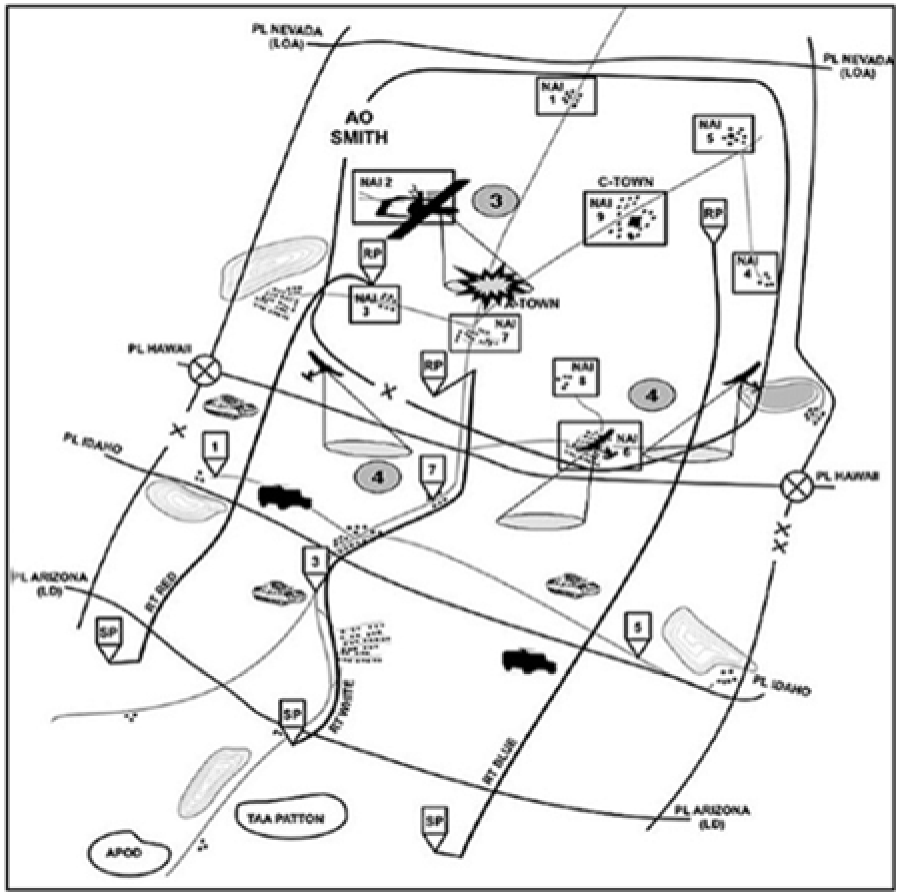
- Zone reconnaissance:
  - Involves a directed effort to obtain detailed information on all routes, obstacles, terrain, and enemy forces within a zone defined by boundaries.



Reconnaissance Types

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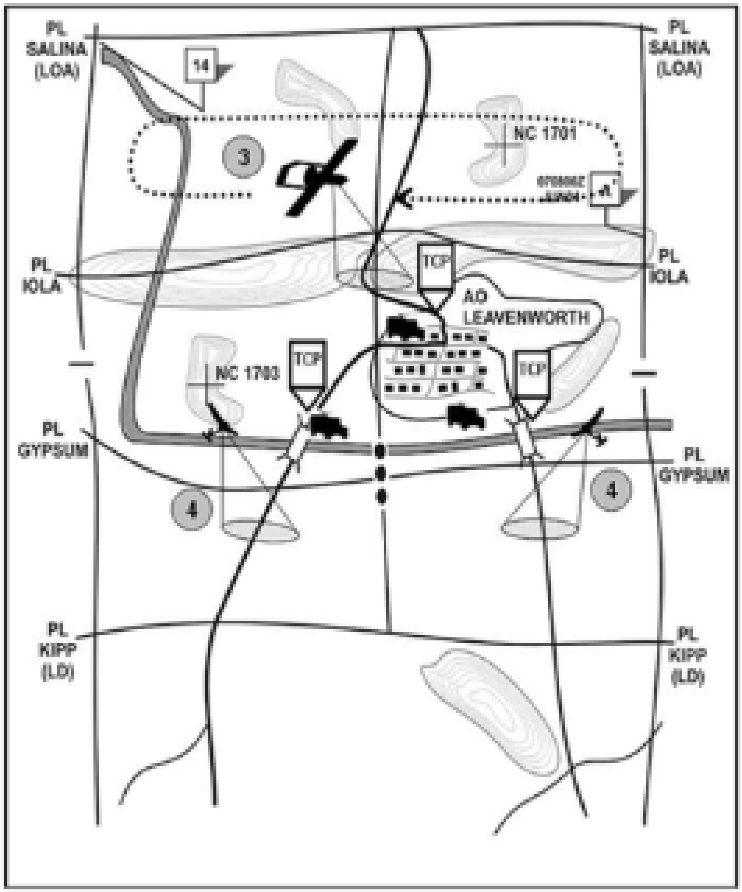
- Area reconnaissance:
  - Focuses on obtaining detailed information about the terrain or enemy activity within a prescribed area.



Reconnaissance Types

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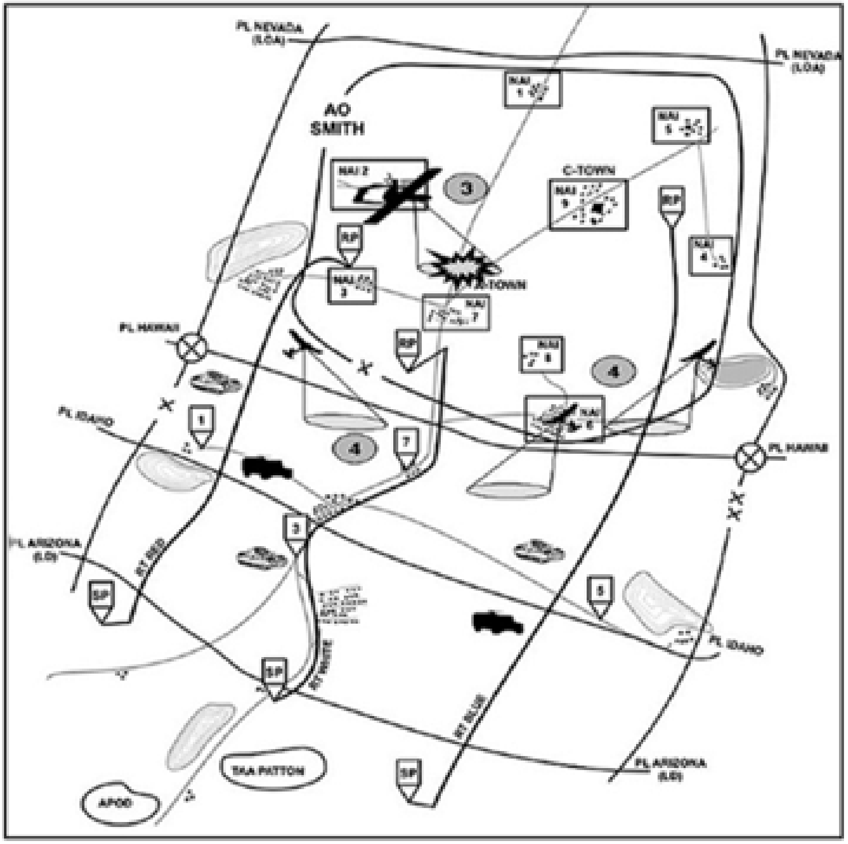
- Route reconnaissance:
  - To obtain detailed information of a specified route and all terrain from which the enemy could influence movement along that route.



Reconnaissance Types

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- Reconnaissance in force:
  - Designed to discover or test the enemy's strength, dispositions, and reactions or to obtain other information.
- Special reconnaissance:
  - Reconnaissance and surveillance actions conducted as a special operation in hostile, denied, or diplomatically and/or politically sensitive environments to collect or verify information of strategic or operational significance, employing military capabilities not normally found in conventional forces.

Reconnaissance Methods

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- Mounted reconnaissance:
  - Enables a more rapid tempo while increasing the potential compromise of reconnaissance efforts. Mounted reconnaissance should take advantage of standoff capabilities provided by surveillance and weapon systems to observe and engage from greater distances.

Reconnaissance Methods

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- Dismounted reconnaissance:
  - Most time-consuming method used by ground units but permits the most detailed information collection about the enemy, terrain, civil considerations, and infrastructure.

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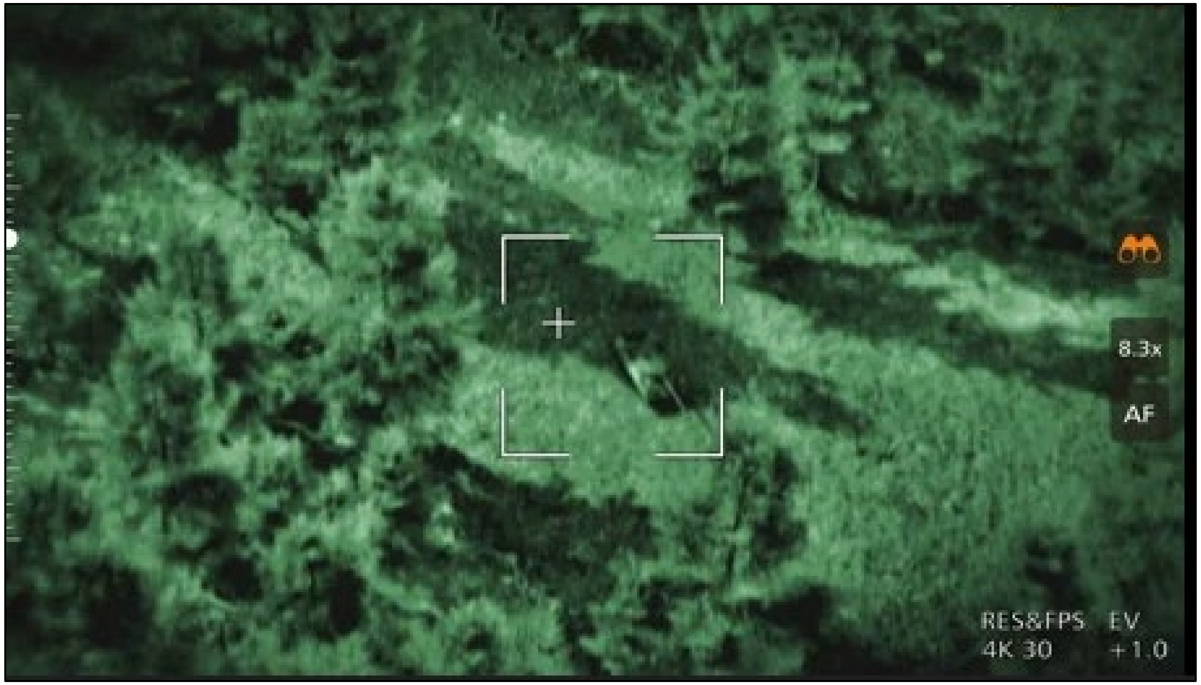
Reconnaissance Methods

- Aerial reconnaissance:
  - Conducted by Army or joint aviation assets that serve as a link between sensors of mounted or dismounted reconnaissance.



Reconnaissance Methods

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- Reconnaissance by fire:
  - A technique in which a unit fires on a suspected enemy position to cause the enemy to react by moving or returning fire, thus disclosing their disposition or willingness to fight.

Reconnaissance vs Surveillance

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- Surveillance is the systematic observation of aerospace, cyberspace, surface, or subsurface areas, places, persons, or things by visual, aural, electronic, photographic, or other means.
- Surveillance involves observing an area to collect information and observing the threat and local population in an NAI or a TAI.
- Surveillance may be a stand-alone mission or part of a reconnaissance mission (particularly area reconnaissance).
  - Elements conducting surveillance must maximize assets, maintain continuous surveillance on all NAIs and TAIs, and report all information accurately.

Reconnaissance vs Surveillance

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- A variety of assets, means and mediums can perform surveillance tasks.

Reconnaissance vs Surveillance

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- Effective surveillance:
  - Maintains continuous observations of all assigned NAIs and TAIs.
  - Provides early warning.
  - Detects, tracks, and assesses key targets.
  - Provides mixed, redundant, and overlapping coverage.
- Surveillance is distinct from reconnaissance in that surveillance is tiered and layered with technical capabilities and systems that collect information.
- Surveillance is passive and continuous; reconnaissance is active in the collection of information and usually includes human participation and fighting for information.

Reconnaissance vs Surveillance

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- Surveillance and reconnaissance complement each other by cueing the commitment of collection capabilities against locations or specially targeted enemy units

MOP

MOE

Priority Intelligence Requirement	Indicators	Specific Information Requirement	NAI	Start	Stop	Assets													Decision Point	Target Area of Interest
						Brigade							EAB							
						1st BN	2d BN	3d BN	3/67 CAV REG	Shadow	Prophet/LLVI	HCT	COMINT	ELINT	HUMINT	GEOINT	CI	MASINT		
1. Where along AA1 will the 375th BTG initiate shaping operations for an area defense?	1. Special purpose forces in hasty battle positions in vicinity EA1 and EA2	1.1.1 Report communications coordinating enemy movement	1,2	H-48	H+2	C	C	C	C	NT	TA	NT	R	R					1	1
		1.1.2 Report movement of fighters into defensive positions	1,2	H-48	H+2	C	C	C	TP	TA	TA	NT	R			R		R	1	1
		1.1.3 Report communications of reconnaissance assets	1,2	H-48	H+2	C	TA	C	TP	TA	TP	NT	R	R					1	1

Reconnaissance Fundamentals

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1. Ensure continuous reconnaissance.
2. Do not keep reconnaissance assets in reserve.
3. Orient on the reconnaissance objective.
4. Report all required information rapidly and accurately.
5. Retain freedom of maneuver.
6. Gain and maintain enemy contact.
7. Develop the situation rapidly.

Ensure Continuous Reconnaissance

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- BCTs require continuous information collection throughout all phases and critical events of all operations.
- Provides commanders with a constant flow of relevant information to identify and seize key terrain, confirm or deny enemy composition, disposition, strength, and courses of action.
- Provides reaction time and maneuver space for unpredicted enemy actions.

Ensure Continuous Reconnaissance (EW)

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- Ensuring you have enough power or batteries to last for the duration of your mission.
- Multiple people assigned to each sensor to ensure you have a work/rest cycle to support 24-hour operations.
- Constantly assessing and refining your observation point to ensure it provides adequate coverage of the objective.

Do not keep Reconnaissance Assets in Reserve

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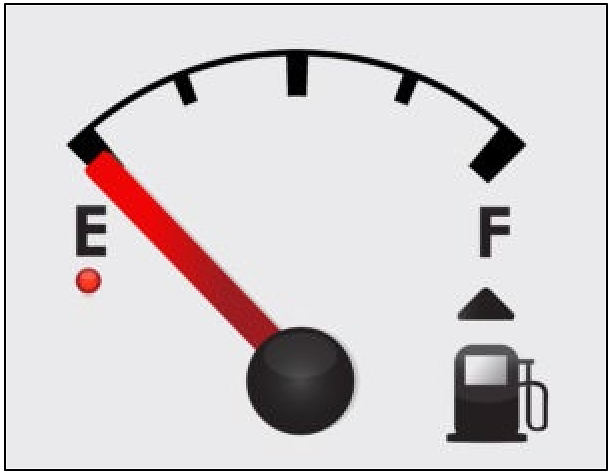
- Commanders maximize employment of their reconnaissance assets and capabilities to answer their CCIRs.
- BCTs task and position reconnaissance assets and capabilities at the appropriate time, place, and in the right combination (human, sensor, and technical means) to maximize their impact, allow for timely analysis of information, and aid decision-making at the appropriate echelon.



Do not keep Reconnaissance Assets in Reserve (EW)

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- Attaching to or supporting other reconnaissance assets near the Forward Line of Own Troops (FLOT).
- Actively making recommendations to CMD/Staff of employment of the EW PLT.
- Avoiding being placed at the BDE Main CP or in the rear, unless mission dictates.

Orient on the Reconnaissance Objective

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- Commanders establish the reconnaissance objective with a specific task, purpose, and focus to direct reconnaissance efforts.
- The reconnaissance objective specifies the most important result to obtain from the reconnaissance effort and clarifies the intent of the reconnaissance effort.

Orient on the Reconnaissance Objective (EW)

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- Identifying the objective in both physical terrain and in the EMS.
- Actively utilizing and refining the Enemy EOB and SOIs to ensure effective surveillance in the EMS.
- Maintaining communication with the Staff, CEMA and the supported unit to maintain situational awareness.

Report all Required Information Rapidly and Accurately

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- Commanders develop plans and make decisions based upon the analysis of information subordinate units collect.
- Commanders require quick and accurate reports to make informed decisions on the proper application of forces.
- Rapid reporting allows the staff maximum time to analyze information and make timely recommendations to the commander.
- Information requirements tied to decision points with a Latest Time Information is Of Value (LTIOV) date-time group provide focus for units collecting information and ensure units report information to facilitate timely decisions.

Report all Required Information Rapidly and Accurately (EW)

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- Identifying and prioritizing information by how it supports the mission.
- Establishing SOPs for PLT internal and external reports.
- Assessing and rehearsing your PLT knowledge management systems.
- Rehearsing your PLT external knowledge management systems.
  - i.e. - Who are you talking to and how do you talk to them?
- Pre-planning communication routes and reports to ensure the correct unit or staff gets the information that they can access, without overloading them with reports.
- Multiple communications avenues to send information.
  - Do not stove pipe information!

Retain Freedom of Maneuver

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- Tactical mobility and maneuver drive the success of reconnaissance operations.
- Reconnaissance operations confirm or deny assumptions about terrain and enemy made during mission analysis and IPOE to identify opportunities and maintain freedom of maneuver for the BCT.
- Counterreconnaissance operations retain freedom of maneuver by denying enemy collection efforts and identifying opportunities for the command to seize, retain, and exploit initiative.

Retain Freedom of Maneuver (EW)

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- Evaluating your position by your ability to move into and out of it.
- Account for your vehicle type and consider its terrain restrictions.
- Plan for alternate observation posts and establish and rehearse criteria for when to move.
- Establish and rehearse criteria for taking and breaking contact.

Gain and Maintain Enemy Contact

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- Using at least one of the nine forms of contact, commanders and staffs plan for and integrate aerial and ground sensors, manned platforms and unmanned systems, dismounted operations, SIGINT, geospatial intelligence, HUMINT, and visual observation to gain contact with the enemy using the smallest element possible.
- Maintaining contact with the enemy provides real-time information of the enemy's composition, disposition, strength, and actions that allow staffs to analyze and make recommendations to the commander.

Gain and Maintain Enemy Contact (EW)

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- Fighting the enemy and not the plan; if you can't make EMS contact where you planed then continue to maneuver until you do.
  - Within reason, don't become compromised or engaged.

Develop the Situation Rapidly

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- Reconnaissance forces act instinctively and urgently to increase the commander's situational understanding of the terrain, enemy, and civilian population.
- Effective reconnaissance forces understand how time impacts movement (both friendly and enemy) and how timely collection of information requirements supports the commander's decision-making.
- The reconnaissance scheme of maneuver and tempo matches the requisite urgency to answer the necessary information requirements.

Develop the Situation Rapidly (EW)

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- Reporting accurately and timely.
  - If you find no enemy EMS contact at a likely enemy position, this could be a significant to report depending on the PIR to be answered.
- Advise and make recommendations to CMD/Staff on the EW mission.
- Understand your CDR's intent and utilize prudent risk to improve and refine your mission within the intent.

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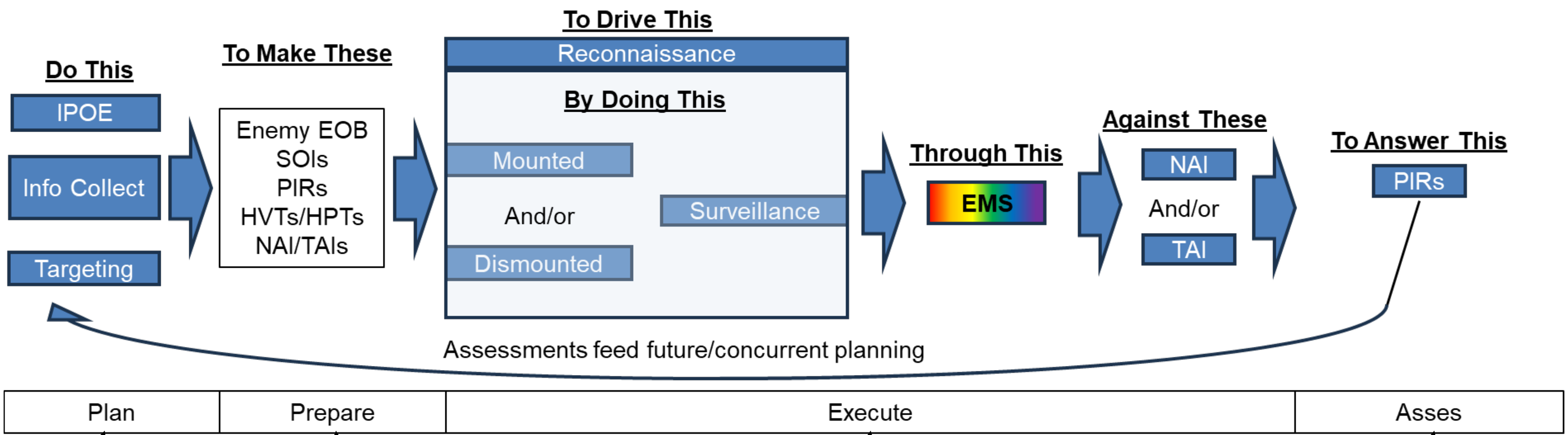
Putting it all together



The EW PLT leadership plans the PLT Mission using TLPs in close coordination with BDE/BN Staffs.

Utilize and refine higher echelon products to prepare their mission.

Execute the mission following the fundamentals of reconnaissance to maneuver to the objective and conduct surveillance by using electromagnetic support TTPs.

Continually assesses the data from EW PLT sensors to answer information requirements and ensures the information the EW PLT provides is timely, accurate, and actionable.

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Identify Key Information Reporting for Electromagnetic Spectrum (EMS) Reconnaissance

Reporting in the EMS

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- There is no standard format for all tactical units for EMS reports.
  - Different units operate in different ways.
- Units must develop SOPs to standardize information.
- EW PLT LDRs must ensure that the PLT and its supported unit have well developed and rehearsed SOPs for reports.
- With out standardized reporting, information can be lost or misinterpreted which could lead to potential mission failure.

Types of Information for Reports

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- Emitter Data: Focuses on externals of the signals such as
  - Channel Carrier Frequency
  - Channel Bandwidth
  - Signal characteristics
- Geolocation Data: Gives information on the emitter's location
  - Estimated location
  - Fix, Cut or LOB?
  - Circle Error of Probability (if known)

Types of Reports

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- Stop Jamming Message
  - To stop jamming, the CEWO submits a stop jamming message.

STOP JAMMING MESSAGE

GENERAL INSTRUCTIONS: Use to terminate a jamming task conducted by an electromagnetic attack (EA) asset.

LINE 1 – DATE & TIME: (Date-time group of when jamming should be terminated).

LINE 2 – UNIT: (Unit supported by jamming mission and is requesting termination).

LINE 3 – FREQUENCY: (Enter the radio frequency being jammed or enter "ALL" if jamming is to stop on all jammed frequencies).

LINE 4 – NARRATIVE: (Any additional information required for clarification).

LINE 5 – AUTHENTICATION: (Message authentication if unit standard operating procedures require authentication).

Types of Reports

U.S. ARMY

- Electromagnetic Warfare Frequency Deconfliction Message
  - The CEWO completes an EW frequency deconfliction message to identify and categorize frequencies to be used by friendly forces and to prevent frequency fratricide.

ELECTROMAGNETIC WARFARE FREQUENCY DECONFLICTION MESSAGE [EWDECONFLICT]

REPORT NUMBER: E010 {USMTF #F402}

GENERAL INSTRUCTIONS: Use to promulgate a list of protected, guarded, and TABOO frequencies to ensure friendly force use of the frequency spectrum without adverse impact from friendly electromagnetic attack.

LINE 1 – DATE AND TIME ______________ (DTG)
LINE 2 – UNIT ______________ (unit making report)
LINE 3 – TYPE ______________ (TABOO, protected, or guarded)
LINE 4 – STATUS ______________ (restricted status of frequency: NEW, CHANGE, CANCEL, OR RENEW)
LINE 5 – FREQUENCY ______________ (frequency/frequencies)
LINE 6 – ON TIME ______________ (start DTG of frequency restriction)
LINE 7 – OFF TIME ______________ (end DTG of frequency restriction)
LINE 8 – LOCATION ______________ (UTM or six-digit grid coordinate with MGRS grid zone designator)

****Repeat lines 3 through 8 as a group to accommodate multiple reports. Assign sequential line numbers to succeeding iterations; for example, first iteration 3 through 9; second iteration 3a through 8a; and so on.**

LINE 9 – NARRATIVE ______________ (free text for additional information required for report clarification)
LINE 10 – AUTHENTICATION ______________ (report authentication)

Legend:

DTG	date-time group	MGRS	Military Grid Reference System
EWDECONFLICT	electromagnetic warfare frequency deconfliction	USMTF	United States message text format

Types of Reports

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- Electromagnetic Warfare Mission Summary
  - The CEWO maintains a record of EW missions.

Electromagnetic Warfare Mission Summary [EWMSNSUM]

REPORT NUMBER: E015 {USMTF # G424}

GENERAL INSTRUCTIONS: Use to summarize significant EW missions and the status of offensive EW assets.

Reference: ATP 3-12.3

LINE 1 – DATE AND TIME ______	(DTG)
LINE 2 – UNIT ______	(Unit making report)
LINE 3 – FROM ______	(Beginning DTG Zulu of period summarized)
LINE 4 – THROUGH ______	Ending DTG Zulu of period summarized)
LINE 5 – COUNTRY ______	(Nationality of the target emitter of concern)
LINE 6 – LOCATION ______	(UTM or six-digit grid coordinate with MGRS grid zone designator)
LINE 7 – EMITTER ______	(Emitter call sign and name or nomenclature)
LINE 8 – FUNCTION ______	(Primary function of target)
LINE 9 – NOTATION ______	(Notation or sorting code)
LINE 10 – SIGNAL ______	(Type of signal of target emitter)
LINE 11 – ON TIME ______	(DTG that planned EA activity was initiated)
LINE 12 – OFF TIME ______	(DTG that planned EA activity was terminated)
LINE 13 – PRIORITY ______	(Relative importance of EA mission)
LINE 14 – TYPE ______	(Type of EA used against the emitter)
LINE 15 – PRIMARY FREQUENCY ______	(Primary frequency of EA target signal)
LINE 16 – SECONDARY FREQUENCY ______	(Secondary frequency of EA target signal)
LINE 17 – LOW FREQUENCY ______	(Lower frequency limit of target equipment class)
LINE 18 – HIGH FREQUENCY ______	(Upper frequency limit of target equipment class)
LINE 19 – BANDWITH ______	(Target frequency bandwidth expressed in MHz)
LINE 20 – PULSE REPETITION ______	(Pulse repetition interval or frequency)
LINE 21 – SYSTEM USED ______	(Name/nomenclature of EW asset used to perform the task)
LINE 22 – OPERATIONAL ______	(Number of units that can perform primary EW mission)
LINE 23 – NONOPERATIONAL ______	(Number of units that cannot perform primary EW mission)
LINE 24 – DESTROYED ______	(Number of units that were destroyed in combat)
LINE 25 – CHAFF ______	(Type of chaff)
LINE 26–LOWER FREQUENCY ______	(Lower frequency of a range of frequencies that was blanked by chaff or the lower EA frequency)
LINE 27 – UPPER FREQUENCY ______	(Upper frequency of a range of frequencies that was blanked by chaff or the lower EA frequency)
LINE 28 – LOW LEVEL ______	(Lower altitude in hundreds of feet of airspace that was blanked by chaff)
LINE 29 – UPPER LEVEL ______	(Upper altitude in hundreds of feet of airspace that was blanked by chaff)
LINE 30 – TECHNIQUE ______	(EA technique employed)
LINE 31 – COUNTRY ______	(Country in which chaff was employed)
LINE 32 – ON TIME ______	(DTG that the chaff drop was initiated)
LINE 33–OFF TIME ______	(DTG that the chaff drop was terminated)
LINE 34 – START LOCATION ______	(Start location of the chaff drop in UTM or six-digit grid coordinate with MGRS grid zone designator)
LINE 35 – STOP LOCATION ______	(Stop location of the chaff drop in UTM or six-digit grid coordinate with grid zone designator)
LINE 36 – NARRATIVE ______	(Free text for additional information required for clarification of report)
LINE 37 – AUTHENTICATION ______	(Report authentication)

LEGEND

DTG	date-time group	MGRS	military grid reference system
EA	electromagnetic attack	MHz	megahertz
EW	electromagnetic warfare	UTM	Universal Transverse Mercator

Types of Reports

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- Electromagnetic Warfare Tasking Message
  - The CEWO uses the EW tasking message format to task an EW asset to provide a requested effect.

Types of Reports (Electromagnetic Warfare Tasking Message)

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ELECTROMAGNETIC WARFARE REQUESTING/TASKING MESSAGE [EWRTM]

REPORT NUMBER: E020 {USMTF #A426}

GENERAL INSTRUCTIONS: Use (1) to task component commanders to perform EW operations to Support the overall EW plan, (2) to support component EW operations, and (3) to request EW support From sources outside their commands.

Reference: ATP 3-12.3

LINE 1 – DATE AND TIME ____	(DTG)
LINE 2 – UNIT ____	(Unit making report)
LINE 3 – EA ____	(Electromagnetic activity)
LINE 4 – TASKED ____	(Designator of tasked unit if the JOC is tasking The unit)
LINE 5 – COUNTRY ____	(Nationality of the target emitter of concern)
LINE 6 – LOCATION ____	(UTM or six-digit grid coordinate with MGRS grid Zone designator)
LINE 7 – EMITTER ____	(Emitter call sign and name or nomenclature)
LINE 8 – FUNCTION ____	(Primary function of target)
LINE 9 – NOTATION ____	(Notation or sorting code)
LINE 10 – SIGNAL ____	(Type of signal of target emitter)
LINE 11 – ON TIME ____	(DTG that planned EA activity was initiated)
LINE 12 – OFF TIME ____	(DTG that planned EA activity will be terminated)
LINE 13 – PRIORITY ____	(Relative importance of EA mission)
LINE 14 – TYPE ____	(Type of EA and technique used against the emitter)
LINE 15 – PRIMARY FREQUENCY ____	(Primary frequency of EA target signal)
LINE 16 – SECONDARY FREQUENCY ____	(Secondary frequency of EA target signal)
LINE 17 – LOW FREQUENCY ____	(Lower frequency limit of EA target class)
LINE 18 – HIGH FREQUENCY ____	(Upper frequency limit of EA target class)
LINE 19 – BANDWIDTH ____	(Target frequency bandwidth Expressed in MHz)
LINE 20 – PULSE REPETITION ____	(Pulse repetition interval or frequency)
LINE 21 – ES ____	(Electromagnetic support)
LINE 22 – COUNTRY ____	(Nationality of the target emitter of concern)
LINE 23 – LOCATION ____	(UTM or six-digit grid coordinate with MGRS grid Zone designator)
LINE 24 – EMITTER ____	(Emitter call sign and name or nomenclature)
LINE 25 – FUNCTION ____	(Primary function of target)
LINE 26 – NOTATION ____	(Notation or sorting code)
LINE 27 – SIGNAL ____	(Type of signal of target emitter)
LINE 28 – PRIMARY FREQUENCY ____	(Primary frequency of ES target signal)
LINE 29 – SECONDARY FREQUENCY ____	(Secondary frequency of ES target signal)
LINE 30 – LOW FREQUENCY ____	(Lower frequency limit of target equipment class)

Legend:

DTG	date-time group	JOC	joint operations center
EA	electromagnetic attack	MGRS	Military Grid Reference System
ES	electromagnetic support	MHz	megahertz
EW	electromagnetic warfare	UTM	Universal Transverse Mercator

LINE 31 – HIGH FREQUENCY ____	(Upper frequency limit of target class)
LINE 32 – BANDWIDTH ____	(Target frequency bandwidth expressed in MHz)
LINE 33 – PULSE REPETITION ____	(Pulse repetition interval or frequency)
LINE 34 – ON TIME ____	(DTG that planned ES activity was initiated)
LINE 35 – OFF TIME ____	(DTG that planned ES activity was terminated)
LINE 36 – ESSENTIAL ____	(EEI category indicator)
LINE 37 – PRIORITY ____	(Relative importance of ES mission)
LINE 38 – CHAFF ____	(Type of chaff)
LINE 39 – LOWER FREQUENCY ____	(Lower frequency of a range of frequencies that is to be blanked by chaff or the lower EA frequency)
LINE 40 – UPPER FREQUENCY ____	(Upper frequency of a range of frequencies that is to be blanked by chaff or the lower EA frequency)
LINE 41 – LOW LEVEL ____	(Lower altitude in hundreds of feet of airspace to be blanked by chaff)
LINE 42 – UPPER LEVEL ____	(Upper altitude in hundreds of feet of airspace to be blanked by chaff)
LINE 43 – TECHNIQUE ____	(EA technique to be employed)
LINE 44 – COUNTRY ____	(Country in which chaff is to be employed)
LINE 45 – ON TIME ____	(DTG that the planned chaff drop will be initiated)
LINE 46 – OFF TIME ____	(DTG that the planned chaff drop will terminate)
LINE 47 – START LOCATION ____	(Start location of the chaff drop in UTM or six-digit Grid coordinate with grid zone designator)
LINE 48 – STOP LOCATION ____	(Stop location of the chaff drop in UTM or six-digit Grid coordinate with grid zone designator)
LINE 49 – NARRATIVE ____	(Free text for additional information required for report clarification)
LINE 50 – AUTHENTICATION ____	(Report authentication)

Legend:

DTG	date-time group	ES	electromagnetic support
EA	electromagnetic attack	MHz	megahertz
EEI	essential elements of information	UTM	Universal Transverse Mercator

Types of Reports

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- Joint Spectrum Interference Resolution (JSIR)
  - Operators report EMI using this report to CEMA/S-6 or the JSIR-Online Portal on SIPR.

Format for Manually Prepared
Joint Spectrum Interference Resolution Report.

Date of Report: When the report was prepared; not when interference occurred.

1. Originator/Report Preparer Information: Identify the person preparing this report to assist with follow-up actions or questions. Include title, name, organization, location, telephone number, and e-mail address in enough detail to allow anyone reading the report to identify the preparer of this report.
2. Organization Experiencing the Interference Information: Identify the organization by name and location; provide a point of contact with first-hand knowledge of the interference. If the report originator or preparer is the same as the unit point of contact, then state "POC same as originator" or words to that effect.
3. Where and when interference occurred:
   a) Date(s): (include entire date range if more than one day).
   b) Time period: (use precise hour and minute of start and end time, if known).
   c) State/country: (provide geographic name).
   d) Location: (briefly describe the location such as, "on a road through a mountain valley…")
   e) Coordinates: (military grid or latitude and longitude).
4. Description of the type of interference: (meaconing, intrusion, jamming, or interference).
5. Description of the system and radio frequency disrupted or degraded: (nomenclature and frequency/frequencies).
6. Impact of interference to the mission: (describe how the interference is affecting the unit's ability to accomplish the mission).
7. Report all local actions and troubleshooting that have been taken to resolve the problem: (attach additional pictures or documents to report troubleshooting) (Identify the steps taken and whether those efforts had any effect on the interference).
8. Type of assistance required: (indicate specific actions the affected unit would like to occur to mitigate the interference).
9. Cause of interference (if known): (identify what caused the interference and how this determination was made).
10. Recommendation for improving resolution techniques or new frequency allocation: (only filled out by the spectrum investigating unit or frequency manager).

Types of Reports

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- Meaconing, Intrusion, Jamming, And Interference Feeder (MIJIFEEDER) Report
  - The MIJIFEEDER report is used for immediate tactical reporting of EMI.

MEACONING, INTRUSION, JAMMING, AND INTERFERENCE FEEDER REPORT [MIJIFEEDER]

REPORT NUMBER: M020 {USMTF #C120}

GENERAL INSTRUCTIONS: Use to share MIJI incidents in a timely manner and to provide for joint exchange of tactical MIJI information including electro-optical interference.

LINE 1 – DATE AND TIME ________________	(DTG)
LINE 2 – UNIT ________________	(unit making report)
LINE 3 – INTERFERENCE ________________	(strength and characteristics)
LINE 4 – LOCATION ________________	(UTM or six-digit grid coordinate with MGRS grid zone designator of incident)
LINE 5 – ON TIME ________________	(start DTG)
LINE 6 – OFF TIME ________________	(end DTG)
LINE 7 – EFFECTS ________________	(operations or equipment affected)
LINE 8 – FREQUENCY ________________	(frequency or frequency range affected)
LINE 9 – NARRATIVE ________________	(free text for additional information required for report clarification)
LINE 10 – AUTHENTICATION ________________	(report authentication)

Legend:

DTG	date-time group
MGRS	Military Grid Reference System
MIJI	meaconing, intrusion, jamming, and interference
MIJIFEEDER	meaconing, intrusion, jamming, and interference feeder
USMTF	United States message text format
UTM	Universal Transverse Mercator

Types of Non-Standardized Reports

U.S. ARMY

- Frequency Azimuth Power (FAP) Report
- Used for quick information on known SOIs.
  - Operator locations are already known by the unit receiving this report.
  - Mostly used for observation post operations.
- Example:
  - **Frequency/Radio ID** – Charlie (C, from Master Rapper Card)
  - **Azimuth** – 121 (Magnetic North)
  - **Power** – 67 (number is understood as (-) dBm)

Types of Non-Standardized Reports

U.S. ARMY

- EW GEO Report –
  - Used for quick information of suspected transmitter locations while the operator is on the move.
  - Example:
    - **Frequency/Radio ID** – Alpha (A, from Master Rapper Card)
    - **# Corresponding intersecting LOBs** – 8 (# > 3)
    - **Suspected Location** – 93714223 (8–10 digit grid [Grid Zone Designator (GZD) and the 100,000-meter square identifier, will be known by the unit receiving the report])

U.S. ARMY

Example Reporting Battle Drill

BATTLE DRILL #1 Receive Possible TGT over PLT NET

Has the order for prosecution with fires come from the PL/PSG?

YES

Is there already a pre-planned mission for the possible TGT that will adequately achieve effects?

YES

Prosecute TGT # that correlates with the mission IAW the PACE communication plan.

NO

NO

Does the report received over the net affect any other BN's within the BDE?

YES

Does another BN Fires Cell have a pre-planned mission in place that would prosecute the TGT?

YES

Push for prosecution of TGT with corresponding BN.

NO

NO

Does Possible TGT location and azimuth coordinate with TLWS/FSCMS?

YES

Push SALT report to corresponding BN Fires Cell if within their AO.

NO

Continue to monitor the net for possible TGT prosecution, build pre-planned TGT with general location of NAI.

Legend:

PL/ PSG: Platoon Leader/ Platoon Sergeant
TGT: Target
IAW: In Accordance With
PACE: Primary Alternate Contingency Emergency
BN: Battalion
BDE: Brigade
TLWS: Target List Worksheet
FSCMS: Fire Support Coordination Measures
NAI: Named Area of Interest
AO: Area of Operations

U.S. ARMY

Discuss how to Gain and Maintain Enemy Contact in the EMS

Gaining and Maintaining contact in the EMS

U.S. ARMY

- The EW PLT is conducting electromagnetic reconnaissance on an NAI.
  - Trigger:
    - Sensor Picks up a signal
  - Action:
    - Send Emitter report to SIGNT team and/or CEMA cell
  - Effect:
    - Initial indicators for staff

Gaining and Maintaining contact in the EMS

U.S. ARMY

- Trigger:
  - Sensor able to geolocate signal
- Action:
  - Send a tip or cue to other ISR platforms for further conformation
- Effect:
  - Improved situation Awareness

Gaining and Maintaining contact in the EMS

U.S. ARMY

- Trigger:
  - Signal is matched to an SOI/Enemy EOB
- Action:
  - Send a SPOT report and/or Fire mission
- Effect:
  - Actionable effects

Gaining and Maintaining contact in the EMS

U.S. ARMY

- Trigger:
  - Signal is matched to unit off of the Enemy SITEMP
- Action:
  - Answer PIRs
- Effect:
  - Enable CDR decision making

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Summary

- Defined the Fundamentals of Reconnaissance
- Identified Key Information Reporting for Electromagnetic Spectrum (EMS) Reconnaissance
- Discussed how to Gain and Maintain Enemy Contact in the EMS

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Questions